

Software engineering

Introduction

Software is a program or a set of programs containing instructions that provide desired functionality.

Engineering is the process of designing and building something that serves a particular purpose and finds a cost-effective solution to problems.

Software engineering is the process of designing, developing, testing and maintaining software. It is a systematic and disciplined approach to software development that aims to create high quality, reliable and maintainable software.

It includes a variety of techniques, tools, and methodologies including requirements analysis, design, testing and maintenance.

Key principles

1. Modularity: Breaking the sw into smaller, reusable components that can be developed and tested independently.

2. Abstraction: Hiding the implementation details of component and exposing only the necessary functionality to other parts of the s/w.
3. encapsulation: wrapping up the data & functions of an object into a single unit and protecting the internal state of an object from external modifications.
4. Maintenance: regularly updating & improving the s/w to fix bugs, add new features.
5. Testing: Verifying that the s/w meets its requirements.

Software engineering is mainly used for large projects based on s/w systems rather than single program or app.

The main goal of s/w engineering is to develop s/w app for improving quality, budget and time efficiency.

Objectives

- (1) Maintainability: It should be feasible for the s/w to evolve to meet changing requirements.
- (2) Reusability: A s/w product has good reusability if the different modules of product can easily be reused to develop new products.
- (3) Portability: In this case, the s/w can be transferred from one computer system to another.
- (4) Interoperability: Capability of 2 or more functional units to process data.

Advantages

- (1) It improves quality of product.
- (2) It increased productivity of product.
- (3) it may reduce the cost.
- (4) it gives better security and scalability.
- (5) it gives customer satisfaction.

Disadvantages

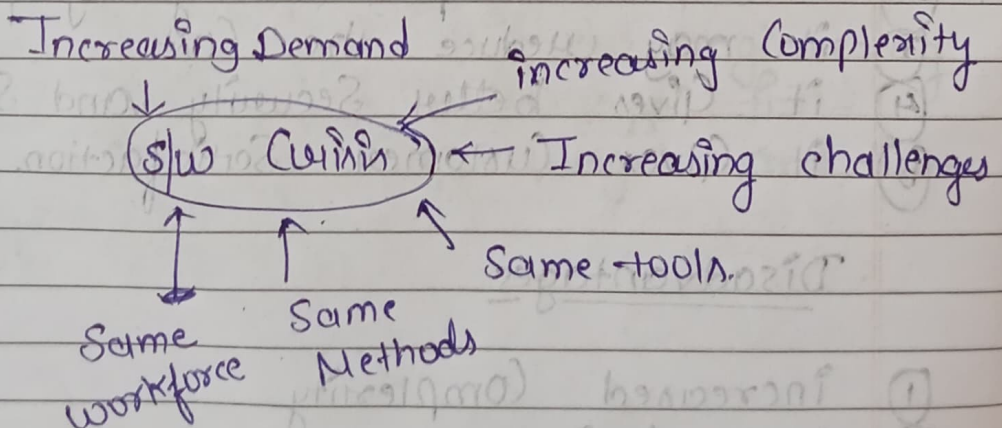
- (1) Increased Complexity
- (2) High Maintenance.

Software Crisis

It is a term used in Computer Science for the difficulty of writing useful and efficient computer programs in the required time.

The SW Crisis was due to using the same workforce, same methods and same tools even though rapidly increasing SW demand, the complexity of SW and SW challenges.

If we will use the same workforce, same methods, and same tools after the fast increase in SW demand, SW complexity and SW challenges then there arise some problems i.e. SW budget problems, SW efficiency problems, SW quality problems, SW management and delivery problem etc. This condition is called a SW Crisis.



Causes

- The Cost of owning & Maintaining s/w was as expensive as developing the s/w.
- At that time s/w was very inefficient.
- The quality of s/w was low quality.
- The s/w Complexity is harder to change.

Solution

There is no single Solⁿ to the Crisis. One Solⁿ is S/w engineering because s/w engineering is a systematic, disciplined and quantifiable approach.

For preventing s/w Crisis, there are some guidelines:

- ① Reduction in s/w Over budget.
- ② The quality of s/w must be high.
- ③ s/w must be delivered
- ④ s/w must meet user requirements.

Program Versus Software

S/w is more than programs. it consists of programs, documentation of program and the procedures used to set up and operate the S/w Systems.

basically,

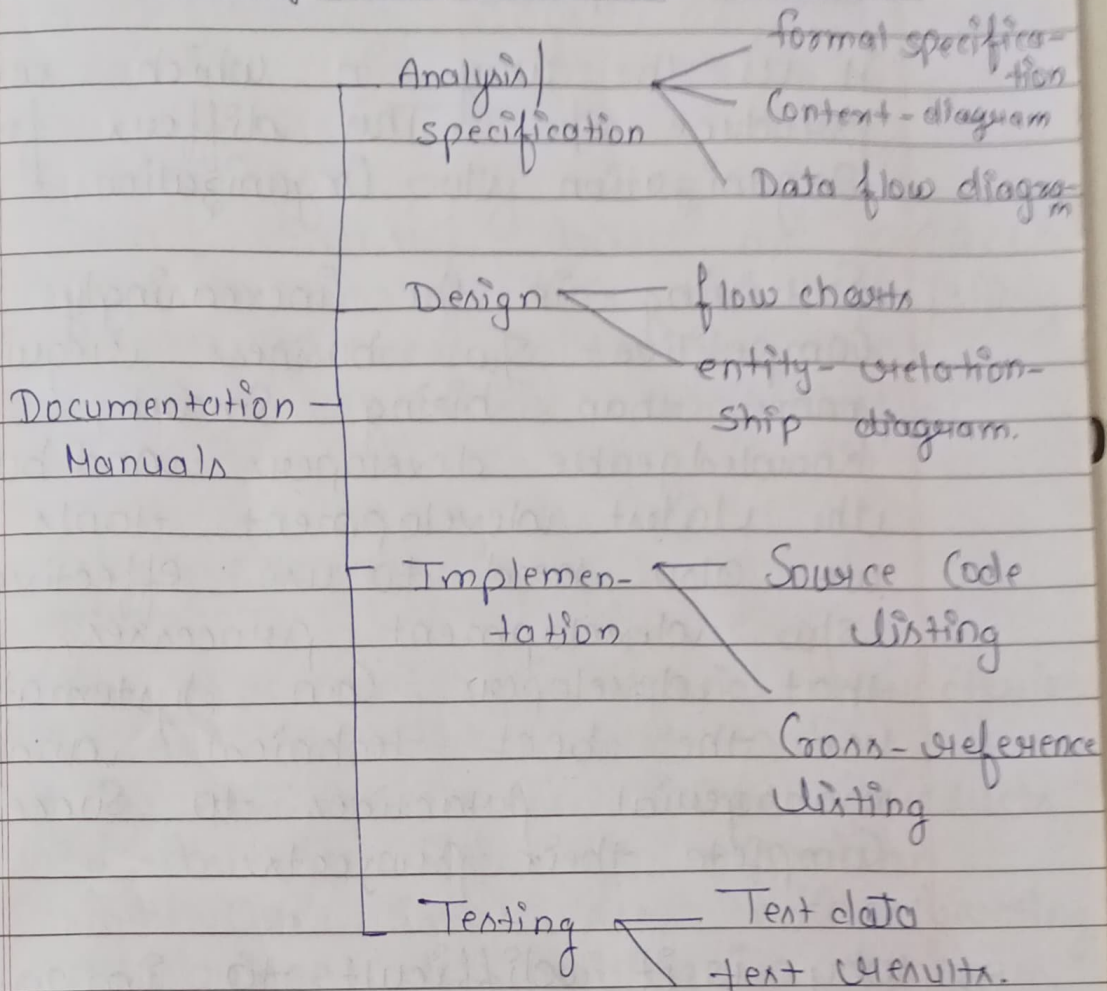
$$\text{S/w} = \text{programs} + \text{documentation} + \text{operating procedures.}$$

Any program is a subset of s/w and it becomes s/w only if documentation and operating procedure manuals are prepared.

Program is a combination of Source Code and Object Code. Documentation consists of different types of Manuals.

operating procedures consist of Instructions to setup and use the s/w system and instructions on how to react the system failure.

List of documentation manuals



List of operating procedure Manuals

